SURROGATE-ASSISTED DOUBLE MACHINE LEARNING: CAUSAL INFERENCE WITH BINARY AND ORDINAL OUTCOMES

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ABSTRACT

Double Machine Learning (DML) delivers root-n consistent, asymptotically normal causal estimates by pairing Neyman-orthogonal scores with cross-fitting, but its partially linear model is confined to continuous outcomes. Binary outcomes are served only by a logit-hardwired logistic estimator; ordinal outcomes by no DML method we are aware of. We introduce SUDO (Surrogate-Assisted DML), which treats the latent outcome behind a categorical response as missing data and fills it in: each observation is completed to a continuous *surrogate* drawn from the assumed error distribution, truncated to the interval the observed category implies, after which ordinary partialling-out applies. The completions are multiple imputations, and pooling them by Rubin's rules is what restores nominal coverage, but only when the draws are *proper*, redrawing the imputation model's parameters (thresholds included) before each draw; naive and improper draws demonstrably under-cover. The method is link-flexible and the surrogate residuals double as a link diagnostic. We further give a closed-form account of how imputation-model error reaches the target: it passes through a truncation factor that rises from about 0.3 to 1 as the signal strengthens, so the imputation model's error is first-order, not the second-order nuisance error of standard DML. In the black-box designs studied, a partially-linear machine-learning imputation model with a full-pipeline bootstrap attains nominal coverage. We illustrate on wine-quality data, estimating the effect of volatile acidity on an ordinal quality score.

1 Introduction

Many outcomes that scientists want causal effects for are categorical. A patient relapses or not; a wine is rated on a five- or ten-point scale; a survey respondent is dissatisfied, neutral, or satisfied. For a *continuous* outcome, Double Machine Learning (DML, Chernozhukov et al. 2018) provides a now-standard route to a causal effect under flexible, high-dimensional adjustment: predict the outcome from the covariates and the treatment from the covariates with any machine-learning models, remove both predictions, and read the effect from the residual-on-residual regression. The removal, a Frisch-Waugh-Lovell (FWL) orthogonalization, is what lets the nuisance predictions be flexible without contaminating the effect, because the score is Neyman-orthogonal and, given standard convergence-rate conditions, is insensitive to first-order nuisance error (Chernozhukov et al. 2018).

That machinery is built for a continuous outcome. A binary or ordinal outcome obeys the additive structure only on a *latent* scale: there is an unobserved continuous utility $U = \theta_0 D + g_0(X) + \varepsilon$, and we see only which interval U fell in. Applying the continuous partialling-out estimator to the raw labels still runs, but it changes the estimand: it returns a probability-scale contrast, not θ_0. An available remedy for the binary case is the logistic partially linear model of M. Liu, Zhang, and Zhou (2021), the DML estimator for a binary outcome with an official implementation in the DoubleML software. It is effective but narrow: the logit link is built into its orthogonal score, so it is confined to binary responses under a logistic distribution and has no ordinal counterpart; for ordinal outcomes we are aware of no DML estimator. Ordinal outcomes are therefore the sharpest gap, and the one this paper closes. Under a logistic link, SUDO targets the same latent-scale coefficient as M. Liu, Zhang, and Zhou (2021) through a different surrogate-completion construction; we do not claim that the two estimators are identical or asymptotically equivalent.

Independently, D. Liu and Zhang (2018) and Cheng, Wang, and Zhang (2021) developed *surrogate residuals* as a diagnostic device for discrete-choice models. Grounded in random utility theory (McFadden 1974), a surrogate maps an observed category to a draw from the conditional distribution of its latent utility, yielding a continuous, homoscedastic residual with a known reference distribution (Greenwell et al. 2018). We repurpose that construction as an *estimation* device. Substituting a surrogate S for the unobserved U turns the discrete problem into a continuous one on which the FWL recipe runs unchanged. The surrogate is a guess, so a single completion is not enough; the completions are multiple imputations, pooled by Rubin's rules (Rubin 1987). Getting that pooling right, imputing from the imputation model's *posterior predictive* rather than a plug-in, is what separates near-nominal coverage from clear under-coverage in every design we study, and is where a naive application fails.

Contributions.

1. A DML estimator for binary and ordinal outcomes in one framework, not tied to the logit link, with a built-in link diagnostic.
2. The finding that only the proper multiple-imputation draw scheme attained nominal coverage in the designs we study, quantified against naive and improper alternatives.
3. A closed-form analysis of how imputation-model error propagates to the effect, revealing a *first-order* sensitivity absent from standard DML, confirmed by a direct perturbation experiment.
4. A route to nominal coverage with black-box machine-learning imputation models, via a partially-linear structure and a full-pipeline bootstrap.
5. An application to wine quality, an ordinal latent-scale estimand for which we are aware of no existing DML estimator.

2 Setup and estimand

Let D be a treatment, $X \in \mathbb{R}^p$ covariates, and Y a categorical outcome: binary, $Y = \mathbf{1}\{U > 0\}$, or ordinal, $Y = j \iff \kappa_{j-1} < U \le \kappa_j$ for thresholds $-\infty = \kappa_0 < \kappa_1 < \cdots < \kappa_{J-1} < \kappa_J = \infty$ (we write κ rather than c to keep the thresholds distinct from the pass-through factor $c(\cdot)$ of Section 13). The latent utility follows a partially linear model

$$U = \theta_0 D + g_0(X) + \varepsilon, \qquad \varepsilon \mid D, X \sim F \tag{1}$$

with g_0 an arbitrary nuisance function and F a known error distribution (logistic, giving proportional-odds; extreme-value, giving complementary-log-log; etc.). The target θ_0 is the treatment effect on the latent-utility scale, the same quantity a correctly specified cumulative-link model targets, but with g_0 left nonparametric. It is not the probability-scale average treatment effect; the two answer different questions and should not be compared against a common truth. Because F fixes the variance of ε, this scale is set by the link rather than estimated, which is what gives θ_0 its familiar reading: under the logistic link a one-unit change in D shifts the cumulative log-odds of Y by θ_0, so $\exp(\theta_0)$ is a proportional-odds ratio.

Four assumptions identify θ_0. (i) *Selection-on-observables*: no unobserved confounder acts beyond X; together with the conditional location-family statement in Equation 1, the latent error has the same normalized law across treatment values given X. (ii) *A constant latent-scale effect*: the partially linear form (Equation 1) makes θ_0 a single number rather than a function of X. (iii) *A correctly specified latent distribution*: the surrogate completion draws from F and the assumed cutpoints, so the conditional law of ε, not merely its mean, must be right; this is the assumption the link diagnostic monitors, and the one the propagation analysis shows the estimate is most sensitive to. (iv) *Overlap*: the treatment keeps support after adjustment, so its residual is not annihilated. Assumptions (i) and (iv) are the standard causal conditions, made explicit for the application through a stated causal graph; (ii) and (iii) are modeling choices shared with any cumulative-link analysis.

3 Background: FWL, DML, and surrogate residuals

The engine underneath DML is the Frisch-Waugh-Lovell (FWL) theorem: in a linear model the coefficient on D equals the coefficient from regressing the Y-residuals on the D-residuals after partialling X out of both. DML makes this practical for high-dimensional X by estimating the two partialling regressions with flexible machine learning and K-fold cross-fitting, and by using an orthogonal score so that first-order error in those estimates does not bias the effect (Chernozhukov et al. 2018).

FWL still runs on a binary or ordinal label, but it targets the wrong thing: the residual of a $0/1$ variable is bounded and heteroscedastic, and regressing it on the treatment residual estimates a probability-scale contrast rather than the latent-scale θ_0. Recovering θ_0 is the obstacle SUDO removes.

The removal uses *surrogate residuals* (D. Liu and Zhang 2018; Cheng, Wang, and Zhang 2021), developed originally to diagnose discrete-choice models. Grounded in random utility theory (McFadden 1974), the construction maps an observed category to a draw from the conditional distribution of its latent utility, producing a continuous residual that is homoscedastic with a known reference distribution (Greenwell et al. 2018). We use it not to diagnose but to *complete* the outcome, so that FWL, and then DML, can run.

4 Surrogate FWL: the idea in its simplest form

The whole method reduces to one move. Fit an imputation model of the outcome on (D, X) to obtain an estimated latent index $\hat{V}_i$; for each observation draw a surrogate S_i from the assumed distribution F, located at $\hat{V}_i$ and truncated to the interval the observed category implies: $(0, \infty)$ if $Y_i = 1$, $(-\infty, 0]$ if $Y_i = 0$, or $(\hat{\kappa}_{Y_i-1}, \hat{\kappa}_{Y_i}]$ in the ordinal case. When the imputation model is correct and its parameters known, the surrogate is drawn from the conditional law of U given (D, X), so $E[S \mid D, X] = E[U \mid D, X]$ (Theorem 1 of Cheng, Wang, and Zhang (2021)) and the completed data satisfy

$$S = \theta_0 D + g_0(X) + \eta, \qquad E[\eta \mid D, X] = 0 \tag{2}$$

with the same g_0 as in Equation 1, so FWL recovers θ_0. This is the exact case. With an estimated index $\hat{V}$ the equality holds only asymptotically under consistent estimation, and approximately otherwise; the propagation analysis below makes the residual error precise. The demonstration that follows fits the index by maximum likelihood, so it sits in the asymptotic regime.

Nothing more is needed to see the idea work. The self-contained example below (base R, no packages) draws a binary outcome from a logistic latent model, fits the imputation model, completes the outcome to a surrogate, and compares three estimates of θ_0: the logistic-regression coefficient on the treatment, the FWL estimate on the *true* (unobserved) utility (the oracle), and the FWL estimate on the *surrogate* completion.

```r
set.seed(1)
n <- 40000; theta <- 1.5
X <- rnorm(n)
D <- rbinom(n, 1, plogis(0.8 * X))        # treatment depends on a covariate
U <- theta * D + X + rlogis(n)            # latent utility (never observed)
Y <- as.integer(U > 0)                    # only the binary label is seen

V  <- predict(glm(Y ~ D + X, binomial))   # fitted latent index
Flo <- plogis(-V)                         # CDF of logistic(V) at the cutpoint 0
p  <- ifelse(Y == 1, Flo + runif(n) * (1 - Flo), runif(n) * Flo)
S  <- V + qlogis(pmin(pmax(p, 1e-9), 1 - 1e-9))   # truncated-logistic surrogate

rD  <- resid(lm(D ~ X))                   # partial the covariate out of D
fwl <- function(z) sum(rD * resid(lm(z ~ X))) / sum(rD^2)
round(c(glm       = unname(coef(glm(Y ~ D + X, binomial))["D"]),
        oracle    = fwl(U),               # FWL on the true utility
        surrogate = fwl(S)), 3)           # FWL on the surrogate completion
```

```
  glm    oracle surrogate
1.497     1.496     1.505
```

All three land on $\theta_0 = 1.5$: the surrogate completion recovers, from the binary labels alone, the coefficient the latent model would have. The identical construction, an ordinal cumulative-link index with interval truncation, works for ordinal outcomes, and the same estimator is provided as a Python package with a parallel set of R scripts. The rest of the paper does two things to this simple estimator: it hands the partialling to machine learning, and it makes the inference honest, since a single completion understates uncertainty.

5 Surrogate-assisted double machine learning

The estimator. Cross-fit the two adjustment nuisances $\hat{m}(X) \approx E[D \mid X]$ and $\hat{\ell}(X) \approx E[S \mid X]$, and form the residual-on-residual estimate

$$\hat{\theta} = \frac{\sum_i (S_i - \hat{\ell}(X_i))(D_i - \hat{m}(X_i))}{\sum_i (D_i - \hat{m}(X_i))^2} \tag{3}$$

The $\hat{\ell}$ regression uses X only: the index carried D's signal into S, and partialling on X alone strips it back out so the residual regression isolates θ_0. The score $\varphi = (S - \ell(X) - \theta(D - m(X)))(D - m(X))$ is Neyman-orthogonal in (ℓ, m): its derivatives in both nuisance directions vanish by the tower property, so first-order error in $\hat{\ell}$ or $\hat{m}$ does not bias $\hat{\theta}$. This protection covers the two *adjustment* nuisances but not the imputation model that produced $\hat{V}$; the next section analyzes the difference.

Proper multiple-imputation inference. A single completion understates uncertainty, so we draw B surrogates and pool with Rubin's rules (Rubin 1987): the estimate is $\bar{\theta} = B^{-1} \sum_b \hat{\theta}^{(b)}$ and the total variance is $T = \bar{W} + (1 + 1/B)B_*$, the mean within-draw sandwich variance plus the inflated between-draw variance, with a Barnard-Rubin small-sample t reference (Barnard and Rubin 1999). What matters is *how* the draws differ. Each surrogate is centered at an estimated index $\hat{V} = x'\hat{\beta}$; that estimate carries sampling error, and reusing one fitted $\hat{\beta}$ across all draws (an *improper* draw) leaves the shared error invisible to the between-draw variance, so T comes up short. A *proper* draw redraws the imputation model's parameters, $\beta^{(b)} \sim N(\hat{\beta}, \widehat{\mathrm{Cov}}(\hat{\beta}))$ (thresholds included in the ordinal case), before each completion, the large-sample normal form of Rubin's posterior-predictive requirement. Table 1 shows the three schemes moving coverage from 0.80 (naive) to 0.93 (improper) to 0.96-0.97 (proper).

T must account for three things that vary from draw to draw: the sampling noise in $\hat{\theta}$ once a completion is fixed (the within-draw $\bar{W}$), the surrogate randomness in S, and the uncertainty in the imputation model itself. Improper draws discard the last of these. The proper draw restores it by sampling $\beta^{(b)} \sim N(\hat{\beta}, \widehat{\mathrm{Cov}}(\hat{\beta}))$, the asymptotic sampling distribution of $\hat{\beta}$. Under standard regularity this normal limit is shared by the parameter's sampling distribution, its flat-prior posterior, and the parametric-bootstrap distribution, so the proper draw is the large-sample form of a parametric bootstrap of the imputation model, read off the covariance rather than recomputed by refitting. It therefore needs a model that reports a usable covariance, and that requirement is exactly what breaks for a black-box learner, taken up in the machine-learning section below.

Ordinal outcomes and the link. The ordinal case is identical once a cumulative-link model supplies the index and thresholds; because the thresholds are parameters too, the proper draw perturbs them jointly with the coefficients, which is why we fit with a model whose covariance covers them. The error distribution F is an assumption, checked by a diagnostic: under a correct link the surrogate residuals $R = S - \hat{V}$ are homoscedastic, so a smooth of R^2 on X (on an independent draw stream) flags a misspecified link through variance drift that mean-based checks miss.

6 How imputation-model error propagates

The orthogonality established above protects $\hat{\theta}$ from error in the adjustment nuisances but says nothing about the imputation model, whose error is baked into the completed data S rather than differenced away. This section quantifies that channel. It is the structural difference between SUDO and standard DML: the imputation model, unlike the two adjustment nuisances, sits outside the orthogonality guarantee, so the method is DML-assisted rather than fully orthogonal, and its overall score is not insensitive to every estimated component.

The argument below needs the fitted index to be free of the observation's own outcome, and two routes deliver that. A *flexible* imputation model is cross-fit, so $\hat{V}_i$ is an out-of-fold prediction and is independent of Y_i given (D, X). A *parametric* one is fit once on all the data, where $\hat{V}_i$ depends on Y_i only at order $1/n$, which is asymptotically negligible. Cross-fitting a parametric imputation model is not merely unnecessary but harmful: the per-fold index noise inflates the between-draw variance, and we measured coverage rising to 1.00 with the mean standard error at 2.3 times the true standard deviation. Accordingly the binary machine-learning design below uses a cross-fit `gam` imputation model, while the ordinal designs and the wine application fit `clm` once.

Either way $\hat{V}$ is fixed with respect to the completion, while $Y \mid D, X$ is generated by the truth. Since S is $\hat{V}$ plus truncated noise,

$$E[S \mid D, X] = \psi(\hat{V}, \eta), \quad \psi(v, e) = v + F(e)\,\mu_+(v) + (1 - F(e))\,\mu_-(v) \tag{4}$$

with $\mu_\pm$ the truncated-noise mean corrections. The correctly centered draw is exact, $\psi(e, e) = e$ (recovering Theorem 1 of Cheng, Wang, and Zhang (2021)), but index error passes through with a factor strictly below one,

$$c(e) = \frac{\partial \psi}{\partial v}\Big|_{v=e} = 1 - f(-e)\,(\mu_+(e) - \mu_-(e)), \qquad f = F' \tag{5}$$

Read c as the fraction of index error that survives the truncation. With no truncation the surrogate would be index plus noise and its mean would move one-for-one with $\hat{V}$, giving $c = 1$; the observed category drags it back, and c is what is left.

Under a logistic link c rises from 0.31 at $e = 0$ to 1 in the tails (Figure 1). The value at the origin is exactly $1 - \log 2$, because for the logit

$$c(e) = 1 - H(F(e)), \qquad H(p) = -p \log p - (1 - p) \log(1 - p), \tag{6}$$

so the self-correction is precisely the entropy of the outcome. That is the whole mechanism in one line: at $p = 1/2$ the label is maximally uncertain, so learning it is maximally informative about where U fell and the correction is largest; as $p \to 1$ the label was already predictable, tells us nothing we did not have from $\hat{V}$, and cannot correct anything. It is also why strong effects are the hard case, and why an ordinal category spanning many error standard deviations behaves badly: both make the observed outcome uninformative about U.

The density in Equation 5 is evaluated at $-e$, which matters for an asymmetric F such as the extreme-value link; writing $f(e)$ is a shorthand valid only when F is symmetric. Section 13 gives the general-J form of both Equation 4 and Equation 5, where the ordinal thresholds contribute their own derivative block. For the well-placed partitions studied here, adding categories lowers the pass-through at $e = 0$: from 0.307 for binary to 0.157 and 0.090 for the displayed three- and four-category configurations. Category count alone is not monotone, however; widely spaced thresholds can instead drive the pass-through toward one.

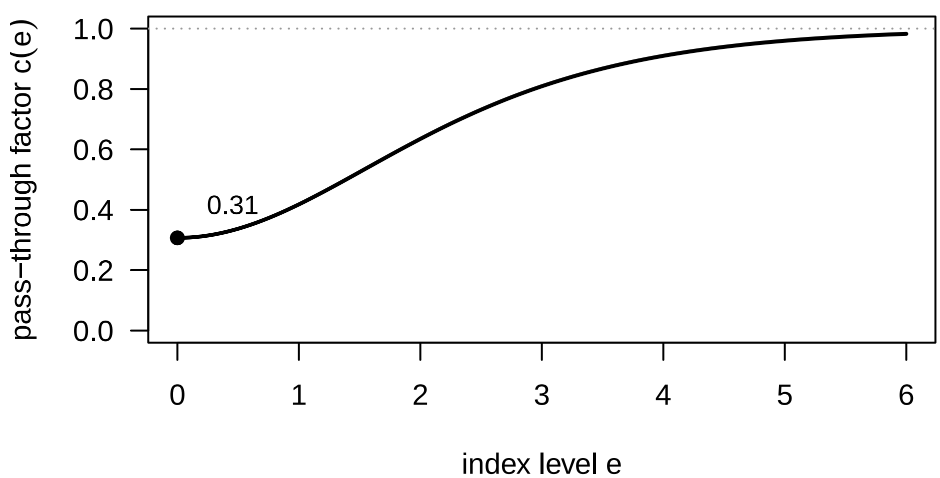

Figure 1: Pass-through factor $c(e)$ under a logistic link. Imputation-model error at signal level e reaches $\hat{\theta}$ multiplied by $c(e)$, which climbs from about 0.31 at $e = 0$ toward 1 as the signal grows: the truncation self-corrects when Y is informative and stops doing so once Y is near-deterministic. Averaging c over the index distribution gives the $\bar{c}_1$ quoted in the text.

Linearizing ψ around the truth, the bias of $\hat{\theta}$ is approximately $\bar{c}_1\,\delta_D$ plus a covariate-direction leak given below, where δ_D is the imputation model's error in its effective treatment coefficient and $\bar{c}_1 = E[w\,c(\theta_0 + g)]/E[w]$, $w = m(1 - m)$.

Three consequences follow. First, the truncation *self-corrects*: because Y comes from the truth, it drags S toward the correct interval, which is why $c < 1$. But $c \to 1$ as the signal grows: once Y is nearly deterministic it adds no information beyond $\hat{V}$, and the self- correction switches off. This, not any numerical instability, is why inference is hardest at strong effect sizes.

Second, the covariate direction is not protected either, which is worth spelling out because the opposite is the natural guess. Split the imputation model's index error into a treatment part and a covariate part, $\widehat{V} - \eta = \delta_D D + \delta_X(X)$. Partialling-out regresses on $R = D - m_0(X)$, and $E[R \mid X] = 0$ annihilates any function of X, so δ_X ought to wash out. It does not, because S is nonlinear in $\widehat{V}$: linearizing multiplies the index error by $c(\eta)$, and η depends on D, so δ_X arrives multiplied by something that itself varies with treatment and therefore survives the projection. For binary D, writing $\eta_d = \theta_0 d + g_0(X)$ and $w = \mathrm{Var}(D \mid X)$, the surviving term is

$$\text{covariate-direction leak} \;=\; \frac{E[w\,(c(\eta_1) - c(\eta_0))\,\delta_X]}{E[w]}. \tag{7}$$

The leak is thus governed by *how differently the truncation damps the two treatment arms*. Were c the same in both, it would vanish exactly and the naive argument would be right.

It is not always small. It does vanish when the imputation model can represent g_0, which is why designs with every model correctly specified show no covariate-direction bias. When it cannot, the leak bites: giving g_0 an X_1X_2 interaction that an additive imputation model cannot express produces a bias of 5 to 6% of θ_0 even though the treatment enters that model correctly, so $\delta_D = 0$ and Equation 7 is the whole story, and coverage falls to 0.75 at strong signal. Refitting the same design with an imputation model that admits the interaction removes the bias, confirming the imputation model rather than the adjustment nuisances as the source.

Third, in the sharpest contrast with standard DML, the imputation model's error enters $\widehat{\theta}$ *linearly*. In DML the outcome is observed and nuisance error enters only through a *product* of two errors (Chernozhukov et al. 2018; Robinson 1988): writing $\ell = \ell_0 + a$ and $m = m_0 + b$ for the errors in the two adjustment models, the remaining bias is $E[ab] - \theta_0 E[b^2]$, which is second order in the pair. Getting the treatment model right ($b = 0$) removes it entirely; getting only the outcome model right does not. The score is Neyman-orthogonal, first-order insensitive to nuisance error, rather than doubly robust in the usual sense. SUDO adds a third model whose treatment-direction error is unprotected and first-order, damped only by $\bar{c}_1 < 1$. Table 3 confirms this numerically: a perturbation applied to the outcome nuisance in DML leaves the estimate unbiased at every level, whereas the same-sized index perturbation in SUDO produces a linear bias with slope exactly $\bar{c}_1$.

7 Machine-learning imputation models

Equation 4 makes imputation-model quality matter more than adjustment-model quality, the reverse of the usual DML intuition. A well-specified parametric imputation model (logistic or cumulative-link) needs no extra machinery, since its covariance supplies the proper draw directly. Such draws propagate parameter uncertainty and attain near-nominal coverage in our parametric experiments; Section 13 characterizes the small Rubin-variance discrepancy that remains. Two obstacles arise with a black-box learner: it can carry a treatment-direction error δ_D (a bias problem), and it has no covariance matrix from which to draw (an inference problem).

The bias is addressed by a *partially-linear* imputation model, $\widehat{V} = \widehat{\beta} D + \widehat{f}(X)$, in which the treatment enters linearly and machine learning handles only X, so no treatment-by-covariate artifact can arise. Fitting $\widehat{f}$ by backfitting a flexible learner is unbiased across effect sizes in our experiments. The inference problem is addressed by a *full-pipeline bootstrap*, the nonparametric counterpart of the proper draw: where a parametric model injects imputation-model uncertainty by sampling β from its covariance, here we resample the whole data set and rerun the entire procedure, cross-fitting and all, using the spread of the results as the standard error. Both target the same variance component; the bootstrap simply obtains it by refitting when no covariance is available to sample from. It has to wrap the *whole* pipeline, because a within-fold refit of the model alone under-propagates that same imputation-model variability. Together these restore nominal coverage with a black-box imputation model (Section 8).

We report the bootstrap as an empirical result rather than a theorem. What would be needed is asymptotic linearity of the imputation-model estimator uniformly over resamples, the Section 13 rate conditions holding conditionally on the data, fold assignment redrawn within each resample, and differentiability of the generated-outcome map. The first three are close to available: Tang and Westling (2024) give conditions for bootstrap consistency of asymptotically linear estimators with machine-learned nuisances, and Lin and Han (2026) establish bootstrap validity for general DML estimators under exchangeably weighted resampling, under the same conditions that make DML itself valid. Neither covers the generated-outcome step, which is the gap.

A Bayesian ensemble is the obvious alternative, since each posterior draw is a complete alternative imputation model and so is a proper draw by construction. The mechanism is even tidier than it first appears: under a probit imputation model the Albert-Chib augmentation variable is exactly the surrogate completion, so a Gibbs sweep returns one for free. We tried it and it did not pay off. On a probit analogue of the machine-learning design, imputation by BART, both unconstrained and with the partially-linear constraint above, carried a bias that grew with signal strength to about 5% of

θ_0, where the backfit is unbiased, and its posterior propagated variance no better than the bootstrap it was meant to replace. That is one design and not a general verdict, but we report it because the elegance argument is the natural thing to try next and it did not survive contact. The run is in the repository. A forest's out-of-bag or infinitesimal-jackknife variance (Wager, Hastie, and Efron 2014) is not such a route at all: it quantifies prediction uncertainty at a point, not a coherent joint draw of the whole imputation model.

8 Simulation study

Every design follows one recipe. Covariates are drawn independently, treatment is assigned from them, a latent utility is formed, and the observed category is read off:

$$X_1, \dots, X_p \overset{\text{iid}}{\sim} N(0,1), \qquad D \mid X \sim \text{Bernoulli}(\Lambda(\pi(X))), \qquad U = \theta_0 D + g_0(X) + \varepsilon, \\ \varepsilon \sim F, \qquad Y = j \iff \kappa_{j-1} < U \le \kappa_j, \tag{8}$$

with Λ the logistic CDF. A design is then fixed by choosing $(p, \pi, g_0, F, \kappa, \theta_0, n)$. Two covariate structures are used throughout: a **one-covariate** setting with $\pi(X) = 0.8X$ and $g_0(X) = X$, where every model in play is correctly specified and the question is purely about pooling; and a **five-covariate machine-learning** setting with

$$\pi(X) = X_4 + \cos X_5, \qquad g_0(X) = X_1^2 + \sin X_2 + 0.5X_3,$$

which is nonlinear in X and makes the treatment depend on covariates that do not enter g_0. Binary outcomes take $\kappa = (0)$; ordinal ones cut U at two interior thresholds into $J = 3$ categories. Table 6 in the appendix lists every design's parameters.

Coverage targets 0.95 and Monte-Carlo error on a coverage figure is about ± 0.02. **SD** is the estimator's true run-to-run variability and **Mean SE** the standard error it reports; Mean SE below SD means intervals that are too short. Learner and tuning specifications, fold and draw counts, and per-cell Monte-Carlo standard errors are in Section 14 and the committed result files, from which the tables below are generated at render time.

Proper draws are what deliver coverage. With a simple binary design and a correct parametric imputation model, only the proper scheme reaches nominal coverage; the naive interval covers 0.80 and the improper 0.93 (Table 1).

Table 1: Inference schemes, binary outcome, $n = 5000$, $\theta_0 = 1.5$, 500 replications. B is the number of surrogate draws.

Scheme	Bias	SD	Mean SE	Coverage
Naive single draw	+0.003	0.083	0.055	0.804
Improper, $B = 25$	+0.004	0.071	0.067	0.924
Proper, $B = 25$	+0.005	0.072	0.076	0.970

Ordinal outcomes. With a three-category outcome and machine-learning adjustment, SUDO is essentially unbiased with valid-to-conservative coverage (Table 2), a regime for which we are aware of no DML competitor. The original implementation was substantially conservative. A term-level ordinal decomposition includes the threshold block and matches Monte Carlo throughout a clean cumulative-logit testbed. It predicts only 2.8% variance overstatement at moderate signal and 2.0% understatement at strong signal, so threshold estimation and ordinal pooling alone cannot explain the 50% to 88% variance gap after flexible partialling nuisances are introduced. A paired nuisance-isolation experiment locates the gap in the fixed outcome nuisance: refitting $E[S \mid X]$ for every ordinal completion reduces mean T by 38% while leaving the empirical sampling variance unchanged, moving T/V from 1.505 to 0.942. The full corrected stage at $B = 25$ and 200 replications has bias 0.016, coverage 0.960, and $T/V = 1.214$ with Monte-Carlo standard error 0.122. Per-draw refitting is therefore the validated ordinal procedure, at the cost of fitting the outcome nuisance B times rather than once. The residual variance ratio is compatible with one at this simulation's precision and is not evidence of a remaining ordinal anomaly. Rubin's variance is guaranteed to err safe only under conditions we have not verified here (a valid imputation model together with a self-efficient complete-data analysis), and under uncongeniality it can be biased in either direction (Meng 1994; Xie and Meng 2017). The over-coverage is a property of these designs, not a general guarantee.

Table 2: Ordinal outcome, $J = 3$, $\theta_0 = 1.0$. Parametric arm 500 replications; corrected spline + ML arm 200, with the outcome nuisance refit per completion.

Configuration	Bias	SD	Mean SE	Coverage
Parametric, $n = 3000$	+0.002	0.077	0.083	0.976
Spline + ML nuisances, $n = 2000$	+0.016	0.104	0.114	0.960

Link misspecification. Generating Gumbel noise but assuming a logistic link biases the latent-scale effect substantially (+0.77 binary, +0.58 ordinal, coverage 0); the correct link removes it (bias < 0.005, nominal coverage), and the variance-drift diagnostic flags the wrong link (p $\approx$ 2e-4) while clearing the right one (p $\approx$ 0.43).

First-order versus second-order sensitivity. Perturbing one input by a known function and tracing the bias (Table 3) confirms the propagation analysis: standard DML with the observed outcome is unbiased under an arbitrarily wrong outcome nuisance provided the propensity is right, whereas SUDO's index error is linear, with treatment-direction slope matching the derived $\bar{c}_1$ (0.669 at $\theta_0 = 1.5$, 0.861 at $\theta_0 = 3$) to within 0.001.

Table 3: Nuisance-sensitivity ladder.

Arm	Perturbed	Bias vs. perturbation
DML	outcome nuisance only	flat, ≈ 0
DML	both nuisances	quadratic
SUDO	index, treatment direction	linear, slope $= \bar{c}_1$

Black-box imputation models. A tuned partially-linear imputation model is unbiased at both effect sizes; pairing it with the full-pipeline bootstrap brings coverage at the strong effect from 0.86 to 0.98 for the normal interval and 0.95 for the percentile interval. The latter is the coverage claim used here. With 100 replications, either coverage estimate has Monte Carlo uncertainty of about ± 0.03; the bootstrap SD also matches the estimator's run-to-run variability at this resolution.

Continuous treatment, and other variations. The designs above all use a binary treatment, while the application below does not, so we check the transfer directly. Replacing binary D with $D = 0.7\pi(X) + N(0, 1)$ in the S5 structure leaves the estimator unbiased for both a binary outcome (-0.1% at $\theta_0 = 1$, $+0.4\%$ at 2.5) and a three-category ordinal one ($+0.7\%$ and $+1.5\%$), so the latent-scale estimand and the machinery carry over. Doubling the covariate count to ten, and scaling the propensity so that overlap is markedly worse, likewise cost nothing in bias. The variation that does matter is the one described above, an imputation model unable to represent g_0. Coverage under a continuous treatment tracks the binary case at moderate signal but is worse at strong signal, where the reported SE falls furthest short of the truth; the full-pipeline bootstrap is the remedy there, as it is for the black-box case. Details and the full grid are in Table 6 and the repository.

9 Application: volatile acidity and wine quality

We estimate the effect of volatile acidity (the acetic-acid "vinegar/nail-polish" fault) on wine quality, an ordinal sensory score (the median of blind tasters) on the UCI wine-quality data (Cortez et al. 2009), 1,599 red and 4,898 white wines. Quality spans six to seven ordinal levels.

This is observational data, so a causal reading rests on an assumed causal structure (Figure 2). Volatile acidity is a plausible treatment because its effect on quality has a clear expected sign (tasters penalize the acetic fault) and because it is a fault endpoint rather than obviously upstream of the other chemistry. A causal reading still rests on strong assumptions: that the remaining physicochemical measurements are pre-treatment adjustments rather than mediators or proxies of volatile acidity, and that winemaking practice has no direct effect on quality beyond the measured chemistry, the exclusion restriction the graph encodes. We do not fully defend these assumptions, but we probe the two most in doubt, robustness to the adjustment set and to unmeasured confounding, below; the application is an illustration on a genuinely ordinal outcome rather than a settled causal finding.

There is adequate residual variation: after adjusting for the other chemistry, volatile acidity retains 75% (red) and 94% (white) of its standard deviation (residual $\sqrt{1 - R^2}$ from the R^2 column of Table 4), so its residual is not annihilated and the residual-on-residual regression is well posed. Applying ordinal SUDO (cumulative-link imputation model, proper draws, cross-fitted nuisances, Rubin pooling) gives a clearly negative effect on latent quality, stable across a linear and a partially-linear (spline) imputation model (Table 4). The naive alternative of reading the coefficient straight

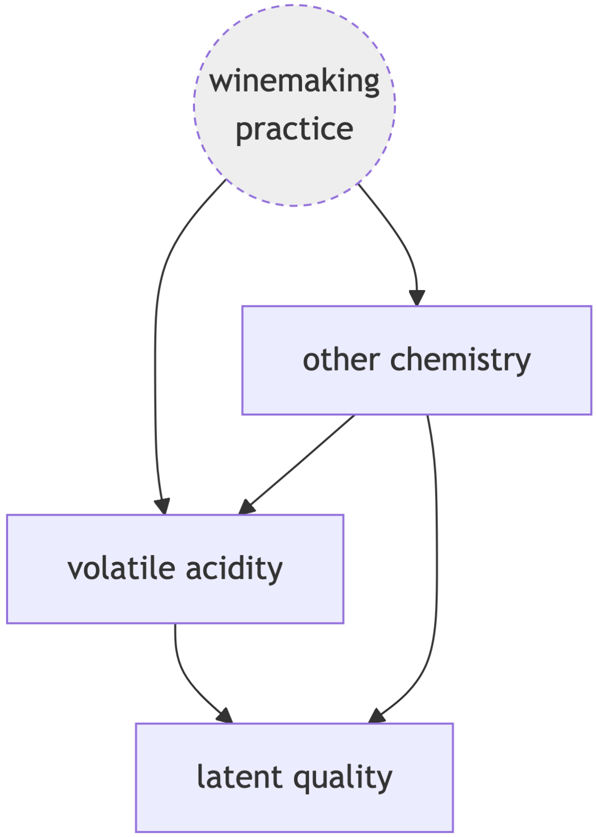

Figure 2: Assumed causal structure. Volatile acidity (the treatment) affects latent quality directly; unobserved winemaking practice confounds both through the measured chemistry, the adjustment set.

off the cumulative-link model reports a much smaller standard error (white: 0.031 versus 0.053-0.074) because it treats the parametric model as exactly correct and ignores the completion and cross-fitting uncertainty SUDO propagates; this is the same inference gap the simulations isolate (Table 1).

Volatile acidity is standardized before fitting, so these coefficients are per one standard deviation (about 0.18 g/dm^3 for red, 0.10 for white), which keeps the effect comparable across the two wines instead of tied to the raw acetic-acid scale. The proportional-odds reading then makes the magnitude concrete: a one-SD rise in volatile acidity multiplies the odds of a wine scoring above any fixed quality level by $\exp(\hat{\theta})$, about 0.59 for red ($\hat{\theta} = -0.53$) and 0.62 to 0.66 for white, a reduction of roughly a third to 40% in those odds per standard deviation.

Table 4: Effect of a one-SD increase in volatile acidity on latent wine quality. Naive read-off from the linear cumulative-link model (standard error in parentheses); SUDO estimates with 95% intervals in brackets.

Wine	Overlap $R^2(D \sim X)$	Naive read-off	SUDO (linear)	SUDO (spline)
Red	0.44	-0.61 (0.07)	-0.53 [$-0.68, -0.38$]	-0.53 [$-0.69, -0.36$]
White	0.12	-0.50 (0.03)	-0.41 [$-0.51, -0.31$]	-0.48 [$-0.63, -0.34$]

The sign is robust. Dropping any single covariate from the adjustment set leaves the effect in $[-0.67, -0.46]$ for red and $[-0.49, -0.37]$ for white, every interval excluding zero, so the estimate does not hinge on one adjustment choice. Read through the omitted-variable-bias framework (Cinelli and Hazlett 2020), an unobserved confounder would have to explain about 16% (red) and 11% (white) of the residual variation in both volatile acidity and latent quality to bring the effect to zero; weaker confounding leaves it negative. We report these robustness values as a heuristic calibration rather than a formal bound: the Cinelli-Hazlett formula is derived for omitted-variable bias in linear regression with an observed outcome, and we are applying it to a pooled, surrogate-completed estimate whose standard error comes from

Rubin's rules. Its validity in that setting has not been established. These values and the leave-one-out ranges are in the committed result files.

How wrong could the imputation model be? Unmeasured confounding is the threat any observational analysis faces; the one specific to SUDO is error in the imputation model, which Equation 7 and Equation 5 show reaches $\widehat{\theta}$ at first order. That channel admits its own sensitivity statement, and unlike the robustness value above it follows directly from the estimator's own theory rather than being imported. Since the bias is $\bar{c}_1\delta_D$, the imputation model's error in the volatile-acidity coefficient that would drive the estimate to zero is $\delta_D = -\widehat{\theta}/\bar{c}_1$. Evaluating $\bar{c}_1$ at the fitted index with the continuous-treatment multiplier $E[c(\eta)DR]/E[R^2]$ and the general-J pass-through of Proposition A1 gives $\bar{c}_1 = 0.21$ (red) and 0.17 (white). The cumulative-link model would therefore have to be wrong about volatile acidity by $+2.55$ and $+2.38$ respectively, which is 4.2 and 4.8 times the coefficient it actually fits. This calculation addresses treatment-direction error only. Conditional on covariate-direction misspecification being negligible, it shows that a large treatment-coefficient error would be required to erase the estimate; it does not establish robustness to general imputation-model error.

Two things make it so, and both are worth noting because they are properties of the design rather than luck. The pass-through is small, near 0.2 rather than the 0.67 to 0.86 of the binary simulations, because a six- or seven-category outcome with well-spaced thresholds constrains the latent utility tightly and the truncation self-corrects hard. And the estimate is far from zero relative to its scale. Both would deteriorate for a coarse outcome or a weak effect, which is exactly when this check is worth running.

10 Discussion

SUDO extends double machine learning to binary and, to our knowledge for the first time, ordinal latent-scale outcomes, by treating the latent outcome as missing data and completing it. Under the logit link it targets the same latent-scale coefficient as the binary-logistic DML estimator of M. Liu, Zhang, and Zhou (2021), while admitting other error distributions and ordinal as well as binary responses. The methodological core is inferential: the completions are multiple imputations, and in our experiments only proper draws, which redraw the imputation model's parameters (thresholds included), attain nominal coverage. The method is link-flexible over any distribution for which the truncated draw and reference are available, and its residuals give a diagnostic that can flag, though not confirm, a wrong link.

The cost, made precise by the propagation analysis, is that the imputation model sits outside the orthogonality guarantee: its error reaches the effect at first order, through a truncation factor that weakens as the signal strengthens. This differs from standard DML, where nuisance error is second-order. It is also manageable: a well-specified parametric or a partially-linear machine-learning imputation model, paired with a full-pipeline bootstrap, attains nominal coverage throughout our experiments.

Several problems remain open, and we are explicit about what Section 13 does and does not settle. Its results are derivations with numerical verification rather than fully discharged theorems: the regularity conditions are stated but not minimized, the treatment-in-imputation-model step still rests on simulation evidence, and the appendix covers only the parametric imputation model fit once on all the data, not the cross-fit flexible regime the machine-learning designs use. SUDO is therefore best read as a simulation-supported estimator with a worked-out first-order theory, not a fully proven inferential procedure. Extending the black-box results beyond the two data-generating processes studied, and to unordered (multinomial) outcomes, which require a vector surrogate and have no single scalar effect, are natural next steps. A Bayesian ensemble imputation model, whose posterior draws are literal proper imputations, is the tidiest candidate for replacing the bootstrap, but a BART version of it did not improve on the bootstrap in the one design we tried; what would make that route work is itself an open question.

11 Data and software

The wine-quality data are from the UCI Machine Learning Repository (Cortez et al. 2009). The estimator is provided as a Python package (`sudo`) together with a parallel set of R scripts that prototype and validate each step; the code and the committed result files behind every table and figure are in the open-source (MIT-licensed) project repository at https://github.com/bgreenwell/sudo.

12 Declaration of AI use

We used Generative AI to assist in writing code and summarizing reference materials. All outputs were carefully reviewed and validated by the authors, who retain full ownership of the final manuscript and accept sole responsibility for any errors.

13 Appendix: asymptotic theory

This appendix collects the theory behind the claims in the body: the pass-through map and its derivatives for general J, the moment properties that separate the adjustment nuisances from the imputation model, the asymptotic distribution of the pooled estimate at fixed B, and the resulting account of Rubin's variance.

A word on status. We label the results propositions and theorems for ease of reference, but they are derivations with numerical verification rather than fully discharged theorems: proofs are given as sketches and the regularity conditions are stated without being minimized. Every closed-form expression below is checked numerically in the project repository, against numerical differentiation for the pass-through derivatives and against Monte Carlo for the variance decomposition, and we flag in place the two points where the argument is incomplete.

13.1 Notation and assumptions

The imputation model is a parametric family $P_\beta(Y \mid D, X)$ with latent index $\eta(D, X; \beta)$; for a GLM with known cutpoints $\eta = x'\beta$. When the cutpoints are estimated, as in a cumulative-link model, we collect them with the index coefficients into a single parameter $\beta = (\beta_\eta, \kappa)$. The thresholds are components of β but not of η: they enter only through the truncation interval, a distinction that matters in Equation 11 below. Write $\hat\beta$ for the maximum-likelihood estimate, $m_0(X) = E[D \mid X]$, $R = D - m_0(X)$, and $J_\theta = E[R^2]$.

Given β, a completion is $S(W; \beta) = \eta(D, X; \beta) + \xi$ with ξ drawn from F truncated to the interval Y implies. The draw noise splits into a data-measurable part and a pure Monte-Carlo part,

$$\xi_i^{(b)} = e_i + u_i^{(b)}, \qquad e_i = E[\xi_i \mid W_i; \beta], \qquad E[u_i^{(b)} \mid W_i] = 0, \tag{9}$$

and this split is what makes the fixed-B bookkeeping come out right. Note that ξ is *not* mean-zero given W: the truncation is selected by the observed Y. Only after averaging over $Y \mid D, X$ at the true model does the mean vanish, because composing $Y \sim P_{\beta_0}$ with the matching truncated draw reproduces the untruncated law, so $\xi \mid D, X \sim F$ exactly. Define

$$\sigma_e^2 = E[e^2 R^2], \quad \sigma_u^2 = E[\mathrm{Var}(\xi \mid W) R^2], \quad \sigma_\varphi^2 = E[\xi^2 R^2] = \sigma_e^2 + \sigma_u^2.$$

We assume throughout: **(A1)** the latent model Equation 1 with selection-on-observables, a constant latent-scale effect, conditional error law $\varepsilon \mid D, X \sim F$, and overlap ($J_\theta > 0$, $\mathrm{Var}(D \mid X)$ bounded away from 0 and ∞); **(A2)** a correctly specified, regular imputation model, so $\eta(D, X; \beta_0) = \theta_0 D + g_0(X)$ and $\sqrt{n}(\hat\beta - \beta_0) \to_d N(0, \Sigma_\beta)$ with $\Sigma_\beta = I(\beta_0)^{-1}$ the inverse per-observation information and influence function $\mathrm{IF}_\beta = I(\beta_0)^{-1} s_\beta(W)$; **(A3)** cross-fitted adjustment nuisances consistent in L_2 with the product rate $\|\hat\ell - \ell_0\|_2 \|\hat m - m_0\|_2 = o_p(n^{-1/2})$, each $o_p(n^{-1/4})$; **(A4)** finite fourth moments and dominated differentiability of $\beta \mapsto E[\varphi]$ at β_0; and **(A5)** proper draws $\beta^{(b)} \sim N(\hat\beta, \hat\Sigma_\beta / n)$, so that $\beta^{(b)} = \hat\beta + n^{-1/2} Z_b$ with $Z_b \mid \text{data} \sim N(0, \Sigma_\beta)$.

The scaling in (A5) is worth stating explicitly, since Σ_β is the limit of $n\widehat{\mathrm{Cov}}(\hat\beta)$ rather than of $\widehat{\mathrm{Cov}}(\hat\beta)$ itself; drawing with the unnormalized matrix would inject variance that does not vanish with n.

Scope. (A2) describes the parametric case: one regular maximum-likelihood fit on all the data, with a covariance to draw from. That is what the ordinal designs and the wine application use, and it is the setting the results below cover. It does not describe a cross-fit flexible imputation model, whose index is assembled from per-fold fits with no single $\hat\beta$ and no covariance of the required form. The binary machine-learning design and the black-box program are therefore outside this appendix, and we report them empirically rather than claiming the theory extends to them.

13.2 The pass-through map for general J

For an outcome in category j the draw truncates ξ to $(a_j, b_j]$ with $a_j = \kappa_{j-1} - v$ and $b_j = \kappa_j - v$ at index v. Write $\mu_j(v, \kappa) = E[\xi \mid a_j < \xi \le b_j]$ and $p_j = F(\kappa_j^0 - e) - F(\kappa_{j-1}^0 - e)$ for the true category probability at true index e. Averaging over $Y \mid D, X$ gives the general form of Equation 4,

$$\psi(v, \kappa; e, \kappa^0) = v + \sum_{j=1}^{J} p_j\, \mu_j(v, \kappa). \tag{10}$$

Proposition A1. *At $(v, \kappa) = (e, \kappa^0)$, writing $t_k = \kappa_k^0 - e$,*

$$c(e) = \frac{\partial \psi}{\partial v} = 1 - \sum_{k=1}^{J-1} f(t_k)\big[\mu_{k+1}(e) - \mu_k(e)\big], \qquad \frac{\partial \psi}{\partial \kappa_k} = f(t_k)\big[\mu_{k+1}(e) - \mu_k(e)\big]. \tag{11}$$

Consequently $c(e) = 1 - \sum_k \partial\psi/\partial\kappa_k$: the index derivative and the threshold derivatives are the same kernel.

Sketch. Differentiate Equation 10. Both truncation limits move with slope -1 in v, and the derivative of a truncated mean with respect to its upper limit is $f(b)(b-\mu)/P$, with the analogous expression at the lower limit. Evaluated at the truth the truncation probability P_j equals p_j, so the weights in Equation 10 cancel the denominators and the surviving terms regroup by threshold. The final identity follows because μ_j depends on (v, κ) only through $\kappa - v$, so shifting the index is the same as shifting every threshold the other way. □

Three consequences. First, the binary case is $J = 2$ with $\kappa_1 = 0$, where the sum collapses to Equation 5. Second, the threshold derivatives are generally nonzero, so for an ordinal outcome both Σ_β and the gradient G of the next section carry a threshold block. This is the formal reason the ordinal full model must be fit with a routine whose covariance covers the thresholds (`ordinal::clm`) rather than one whose does not (`mgcv::ocat`). Third, what governs the pass-through is how much the categorization tells you about U, which is *not* the same as the number of categories. Refining a partition with well-placed thresholds does lower c (at $e = 0$ under a logit, from 0.307 binary to 0.157 and 0.090 for the three- and four-category configurations of Section 15), but widening the same three-category partition reverses this completely, taking $c(0)$ above the binary value and on toward 1. The reason is visible in Equation 11: a category spanning many error standard deviations has a nearly unbounded truncation interval, its μ_j barely moves with the index, and the self-correction switches off. We therefore claim no monotonicity in J. The useful statement is that $c \to 1$ as the observed category stops constraining U, which is the same mechanism that drives $c \to 1$ at strong signal.

The binary logit closed form. For $J = 2$ under a logistic F, the truncated means have an exact expression that turns Equation 5 into Equation 6. Substituting $u = F(x)$, so that $x = \log\{u/(1-u)\}$ and $\mathrm{d}u = f(x)\,\mathrm{d}x$, the partial mean is

$$\int_{-\infty}^{t} x f(x)\,\mathrm{d}x = \int_{0}^{F(t)} \log\frac{u}{1-u}\,\mathrm{d}u = q\log q + (1-q)\log(1-q) = -H(q), \qquad q = F(t),$$

so the partial mean of a logistic is minus the entropy at the corresponding probability. Taking $t = -e$ and $p = F(e)$, and using $H(1-p) = H(p)$ with the fact that the untruncated mean is zero,

$$\mu_-(e) = \frac{-H(p)}{1-p}, \qquad \mu_+(e) = \frac{H(p)}{p}, \qquad \mu_+ - \mu_- = \frac{H(p)}{p(1-p)}.$$

Since $f(-e) = f(e) = p(1-p)$ for the logistic, the factor cancels exactly in Equation 5 and $c(e) = 1 - H(p)$, giving $1 - \log 2 = 0.307$ at $e = 0$. The cancellation is what makes the self-correction *equal* to the outcome's entropy rather than merely proportional to something like it.

13.3 Moment properties

Proposition A2. *Under (A1)-(A2), $E[\varphi] = 0$ at the truth and the moment is Neyman-orthogonal in the adjustment nuisances, $\partial_\ell E[\varphi] = \partial_m E[\varphi] = 0$. It is not orthogonal in β: $\partial_\beta E[\varphi] = G$ with blocks*

$$G_{\beta_\eta} = E\big[c(\eta_0)\,\partial_{\beta_\eta}\eta\, R\big], \qquad G_{\kappa_k} = E\big[f(\kappa_k^0 - \eta_0)\,(\mu_{k+1} - \mu_k)\,R\big].$$

Sketch. At the truth $S - \ell_0 - \theta_0 R = \xi$, and averaging over $Y \mid D, X$ makes $\xi \mid D, X \sim F$, so $E[\varphi] = E[\xi R] = 0$. The mean-zero statement holds only after that averaging; given (W, β) the draw has mean $e \neq 0$. The nuisance derivatives vanish by the tower property. The β derivative is Equation 11 composed with $\partial_\beta \eta$. □

Because any function of X is annihilated by R, profiling ℓ out of the moment leaves G unchanged. That covers either implementation: refitting $\hat{\ell}$ on every completion, or fitting it once from a plug-in completion and holding it fixed, which is what the estimator actually does (see the appendix section on nuisances and draws). Proposition A2 is the formal content of the statement in the body that the imputation model sits outside the orthogonality guarantee.

13.4 Asymptotic distribution at fixed B

Theorem A1. *Under (A1)-(A5), with $\bar{\theta} = B^{-1}\sum_b \hat{\theta}^{(b)}$,*

$$\sqrt{n}(\bar{\theta} - \theta_0) \to_d N(0, V_B), \qquad V_B = J_\theta^{-2}\Big[\sigma_e^2 + G'\Sigma_\beta G + 2G'C + \frac{G'\Sigma_\beta G + \sigma_u^2}{B}\Big], \tag{12}$$

where $C = \mathrm{Cov}(eR, \mathrm{IF}_\beta)$.

Sketch. The Z-estimator expansion gives $\sqrt{n}(\hat\theta - \theta_0) = J_\theta^{-1}\sqrt{n}M_n(\theta_0, \cdot) + o_p(1)$, the sign following from $\partial_\theta E[\varphi] = -J_\theta$. Expanding around (β_0, ℓ_0, m_0), the adjustment-nuisance terms are $o_p(1)$ by Proposition A2 plus cross-fitting plus the (A3) product rate, as in Chernozhukov et al. (2018). The β term is $G'\sqrt{n}(\beta^{(b)} - \beta_0) + o_p(1)$, and $\beta^{(b)} - \beta_0 = (\hat\beta - \beta_0) + n^{-1/2}Z_b$ by (A5). Applying Equation 9 separates the data-measurable score e_iR_i from the draw noise; averaging over b leaves parameter-draw and surrogate-draw terms with conditional variances Σ_β/B and σ_u^2/B. $\square$

Two features of Equation 12 deserve emphasis. First, this is not the single-completion result plus an $O(1/B)$ correction: the leading term itself depends on B. A single completion carries the full sandwich $\sigma_e^2 + \sigma_u^2$ *and* twice the parameter contribution, once from $\hat\beta$'s own sampling error and once from the single draw around it, which is exactly what Rubin's $(1 + 1/B)$ factor exists to undo. As $B \to \infty$ the draw noise averages away and the leading term is σ_e^2, not $\sigma_e^2 + \sigma_u^2$. Second, the cross term $2G'C$ is not sign-restricted, since IF_β and the moment are built from the same data. So V_B is not the DML sandwich plus a nonnegative imputation penalty; it is the sandwich with the draw noise removed by pooling and $G'\Sigma_\beta G + 2G'C$ added.

13.5 The pooled variance

Rubin's total is $T = \bar W + (1 + 1/B)B_*$. The two pieces behave differently, and the distinction matters. $\bar W$ averages a sandwich estimate over n observations, so $n\bar W \to_p J_\theta^{-2}(\sigma_e^2 + \sigma_u^2)$. B_* is the sample variance of B draws. Writing $V_{\mathrm{draw}} = G'\Sigma_\beta G + \sigma_u^2$ for the per-draw variance, Equation 12 gives

$$nB_* \to_d J_\theta^{-2}\, V_{\mathrm{draw}}\, \frac{\chi^2_{B-1}}{B-1} \qquad (n \to \infty,\ B \text{ fixed}), \tag{13}$$

which is *not* a probability limit: at fixed B the between-draw term keeps sample-variance randomness however large n becomes. It is conditionally unbiased for $J_\theta^{-2}V_{\mathrm{draw}}$, which is what lets the comparison below be stated in expectation.

Write $E_\Delta[\cdot] = E[\cdot \mid W_1, \dots, W_n]$ for the expectation over the imputation draws with the data held fixed, so that $E_\Delta[T]$ is still a random variable through the data.

Theorem A2. *Under (A1)-(A5), for fixed B,*

$$n\, E_\Delta[T] - V_B \ \to_p\ 2J_\theta^{-2}(\sigma_u^2 - G'C),$$

with nT itself converging in distribution, not in probability, by Equation 13. Taking expectations over the data as well gives the same limit for the unconditional $n\,E[T] - V_B$, which is the form the Monte-Carlo check below estimates, since each replication redraws both the sample and the imputations. As $B \to \infty$ the randomness in B_* averages away and the statement strengthens to a probability limit for T itself,

$$nT - V_\infty \ \to_p\ 2J_\theta^{-2}(\sigma_u^2 - G'C), \qquad V_\infty = J_\theta^{-2}(\sigma_e^2 + G'\Sigma_\beta G + 2G'C).$$

Either way the $1/B$ terms cancel, so the discrepancy is free of B, and Rubin's variance is asymptotically exact if and only if $G'C = \sigma_u^2$, conservative when $G'C < \sigma_u^2$, and anti-conservative when $G'C > \sigma_u^2$.

Remark (what fixed B costs). With B fixed, no amount of data removes the χ^2_{B-1} fluctuation in the reported variance. This establishes that *some* non-normal reference distribution is necessary at any n, not merely as a small-sample refinement. It does not by itself establish that the Barnard-Rubin degrees of freedom are the right one: that formula is derived for a complete-data analysis whose within- and between-imputation components are independent, whereas here $\bar W$ and B_* are built from the same draws and the same $\hat\beta$. Establishing the required dependence structure is open. The predicted spread is large and is confirmed numerically in Section 15.

Neither direction is automatic, and we do not find exactness. On a deliberately congenial testbed that isolates the pooling from every other source of error (a correctly specified logistic imputation model, continuous treatment so both adjustment nuisances are exactly linear), all six quantities are available in closed form:

θ_0	σ_e^2	σ_u^2	$G'\Sigma_\beta G$	$G'C$	$\sigma_u^2 - G'C$	T/V at $B = 20$
1	1.269	2.021	4.077	1.954	+0.067	1.014
3	0.580	2.710	41.414	3.094	−0.384	0.985

The sign of the discrepancy flips with signal strength: on average T overstates the sampling variance by about 1.4% at $\theta_0 = 1$ and understates it by about 1.5% at $\theta_0 = 3$. The T/V column compares $n\,E[T]$ with V_B, which is the quantity

Theorem A2 governs at fixed B; individual replications scatter around it according to Equation 13. Both discrepancies are asymptotic rather than finite-sample effects; they persist at $n = 20{,}000$ and, being free of B, do not shrink as B grows. The mechanism is visible in the table. As the signal grows the observed Y carries less information beyond the index, so σ_e^2 falls while σ_u^2 and $G'C$ rise, and $G'C$ overtakes σ_u^2. This is a more precise account of why inference is hardest at strong effect sizes than the informal argument in the body, and it corrects a natural but mistaken reading: $c \to 1$ is the regime in which ψ is *closest* to the identity, so the linearization is at its most accurate there, and cannot be what degrades.

The practical import is small: a 1.5% error in a variance is roughly 0.7% in a standard error, well inside the Monte-Carlo error of the coverage figures in Section 8. The theoretical import is that "proper draws plus Rubin's rules are exact here" is not a claim we can make. Section 15 reports the Monte-Carlo check of Equation 12 term by term.

Corollary (improper draws under-cover). If $\beta^{(b)} \equiv \hat{\beta}$, the between-draw term loses the parameter contribution entirely, so B_* is conditionally unbiased for $J_\theta^{-2}\sigma_u^2$ alone and $n\,E[T]$ falls short of V_B. The deficit is dominated by the imputation channel $G'\Sigma_\beta G$ whenever $G \neq 0$ and the signal is not weak, though it is not exactly that quantity, since the cross term and the draw noise also enter. This is the 0.93 versus 0.97 coverage gap in Table 1.

13.6 Bias under a misspecified imputation model

Proposition A3. *Drop the correct-specification half of (A2) and suppose $\hat{\beta} \to_p \beta_*$. Restrict to binary D and to a pure treatment-coefficient error, so that the index error $\eta(\cdot;\beta_*) - \eta_0$ equals δ_D in the treated arm and 0 in the control arm. Then*

$$\bar{\theta} - \theta_0 = \bar{c}_1\delta_D + \textit{(covariate-direction leak)} + o_p(1), \qquad \bar{c}_1 = \frac{E[w\,c(\theta_0 + g_0)]}{E[w]}, \quad w = \mathrm{Var}(D \mid X).$$

Sketch. Linearize $E[S \mid D, X; \beta_*] \approx \eta_0 + c(\eta_0)(\eta(\cdot;\beta_*) - \eta_0)$ and push through $J_\theta^{-1}E[\,\cdot\,R]$. Conditioning on X with binary D, the moment weights the treated arm by w and the control arm by $-w$; the control-arm error is zero by assumption, so only $c(\theta_0 + g_0)$ survives, and $J_\theta = E[w]$ supplies the denominator. □

Both restrictions matter. The displayed $\bar{c}_1$ evaluates c on the treated arm only, which is correct precisely because the control-arm error was set to zero; a general perturbation weights both arms as $m\,c(\theta_0 + g_0) + (1-m)\,c(g_0)$. For continuous D with index error $\delta_D D$ the multiplier is instead $\bar{c}_1 = E[c(\eta_0)DR]/E[R^2]$, which is a different expression. The covariate-direction leak is Equation 7, nonzero because c depends on D through η and so is not annihilated by projecting on R. We do not bound it. It is not negligible in general: with an imputation model unable to represent g_0 it reaches 5 to 6% of θ_0 while $\delta_D = 0$, so nothing here guarantees it is small relative to $\bar{c}_1\delta_D$. Table 3 is the empirical counterpart for the treatment direction, recovering the predicted slope $\bar{c}_1$.

14 Appendix: simulation and application details

Every design draws a latent utility $U = \theta_0 D + g_0(X) + \varepsilon$ with logistic ε and reads off the observed category; the designs differ in the covariate structure, the cutpoints, and the models fit. Covariates are standard normal throughout. Full data-generating code and the committed per-cell result files are in `R/` and `R/results/`.

14.1 Data-generating processes

All designs instantiate Equation 8 with one of two covariate structures:

- **S1**, one covariate: $p = 1$, $\pi(X) = 0.8X$, $g_0(X) = X$. Every model in play is correctly specified, so only the pooling is under test.
- **S5**, five covariates: $p = 5$, $\pi(X) = X_4 + \cos X_5$, $g_0(X) = X_1^2 + \sin X_2 + 0.5X_3$. Nonlinear in X, and the treatment depends on covariates absent from g_0.

Table 6: Data-generating processes. κ lists the interior cutpoints, so (0) is binary and a pair gives $J = 3$.

design	structure	F	κ	θ_0	n	reps
Inference ladder (Table 1)	S1	logistic	(0)	1.5	5000	500
Ordinal parametric (Table 2, top)	S1	logistic	$(-1, 1)$	1.0	3000	500

design	structure	F	κ	θ_0	n	reps
Binary ML	S5	logistic	(0)	0.5, 3	1000, 2000	200
Ordinal ML (Table 2, bottom)	S5	logistic	$(0, 2)$	1.0	1000, 2000	200
Link misspecification	S1	Gumbel	(0) or $(0, 2)$	1.0	5000	200
Sensitivity (Table 3)	S5	logistic	(0)	1.5, 3	2000	200
Black-box imputation	S5	logistic	(0)	1.5, 3	2000	100
Bayesian imputation (repository only)	S5′	normal	(0)	1.5, 3	2000	200
DGP variation, continuous D	S5, $D = 0.7\pi(X) + N(0,1)$	logistic	(0) or $(0, 2)$	1, 2.5	2000	200
DGP variation, interaction	S5 plus X_1X_2 in g_0	logistic	(0)	1, 2.5	2000	200
DGP variation, dimension	S5 with $p = 10$	logistic	(0)	1, 2.5	2000	200
DGP variation, overlap	S5, $\pi(X)$ scaled $2.5\times$	logistic	(0)	1, 2.5	2000	200

S5′ is the probit analogue of S5 used for the Bayesian imputation model, whose binary machinery is probit: the systematic part is divided by $\mathrm{sd}(\text{logistic}) = \pi/\sqrt{3}$ before normal errors are added, which matches the event rates of S5 to within 0.006 and puts θ_0 on the probit scale at $\theta_0/1.8138$. The θ_0 column reports the logit-scale values for comparability. That design supports no table here; it is reported so the claim in the machine-learning section is checkable.

Models fitted. The ladder uses $\mathrm{glm}(Y \sim D + X)$ as the imputation model with $B \in \{10, 25, 50\}$ draws (the table shows $B = 25$). The ordinal parametric design uses `ordinal::clm`, whose covariance covers the thresholds, with $B = 25$. The ordinal ML design uses `clm` on D and natural-spline bases $\mathrm{ns}(X_k, 4)$ with `mgcv::gam` partialling nuisances and $B = 25$.

Link misspecification. The one-covariate structure with extreme-value ε rather than logistic: Gumbel-max for the binary case, matching `glm` with a cloglog link, and Gumbel-min for the ordinal case, matching `ordinal::clm` with a cloglog link. The analyst fits either logit (misspecified) or cloglog (correct). Note this design deliberately keeps the one-covariate structure, so that the only thing wrong is the link.

Sensitivity ladder (Table 3). Starts from the oracle index and oracle partialling nuisances, then adds a known perturbation of size δ to one input: the outcome nuisance only (DML-1), both nuisances (DML-2), the index in the covariate direction (SUDO-X), or the index in the treatment direction (SUDO-D). Bias is traced against a common latent-scale RMSE so the four are comparable. The fitted SUDO-D slope reproduces $\bar{c}_1$ (0.669 at $\theta_0 = 1.5$, 0.861 at $\theta_0 = 3$) to within 0.001, which is the independent check on Equation 5.

Black-box imputation model. Imputation model is a partially-linear backfit $\hat{V} = \hat{\beta}D + \hat{f}(X)$ with $\hat{f}$ a single-hidden-layer neural net (`nnet`, `size = 4`, `decay = 0.3`, five backfitting iterations, selected on theta-level MC bias). Standard errors from the full-pipeline bootstrap with 100 data set resamples.

Wine application. Full model `ordinal::clm` on D and either the raw covariates (linear) or $\mathrm{ns}(X_k, 3)$ (spline); partialling nuisances $E[S \mid X]$ and $E[D \mid X]$ by `mgcv::gam` (Gaussian for the continuous D); five-fold cross-fitting; $B = 50$ proper draws; Rubin pooling. Volatile acidity is standardized.

14.2 Nuisances, cross-fitting, and draws

Partialling nuisances $\hat{\ell}(X) = E[S \mid X]$ and $\hat{m}(X) = E[D \mid X]$ are `mgcv::gam` throughout, fit with five-fold cross-fitting. Proper draws sample the full-model parameters from $N(\hat{\beta}, \widehat{\mathrm{Cov}}(\hat{\beta}))$ (thresholds included for `clm`) before each completion. Diagnostic draws use an RNG stream independent of the estimation draws.

One algorithmic detail matters enough to state. The surrogate S changes with every draw, so its nuisance $\hat{\ell}$ can be refit each time. Earlier binary experiments found refitting negligible, so the binary estimator fits $\hat{\ell}$ once on a plug-in completion and reuses it. The ordinal nuisance-isolation ladder finds the opposite: holding $\hat{\ell}$ fixed nearly doubles the within-draw component and accounts for the stage-5 conservatism. The corrected ordinal estimator therefore cross-fits $\hat{\ell}$ separately on every completion. The full $B = 25$, 200-replication validation attains 0.960 coverage. The correction costs roughly B times as much for the outcome nuisance, which is why the cheaper binary behavior is retained where it has been validated.

14.3 Monte-Carlo precision

Coverage is a binomial proportion, so its Monte-Carlo standard error at a true 0.95 is about $\sqrt{0.95 \cdot 0.05/R}$: 0.010 at $R = 500$ replications, 0.015 at $R = 200$, and 0.022 at $R = 100$. Per-cell bias and coverage Monte-Carlo standard errors are in the committed result files.

15 Supplement: numerical verification of the theory

Every closed-form expression in Section 13 is checked against an independent numerical route. Both scripts are in the project repository and assert their own pass conditions.

15.1 Pass-through derivatives (Proposition A1)

`R/theory_ordinal_passthrough.R` computes ψ by quadrature and compares the analytic index and threshold derivatives of Equation 11 against central differences, for $J = 2, 3, 4$ under logit, probit, and both Gumbel conventions. Maximum discrepancy across all configurations is 3×10^{-10} for the index block and 4×10^{-10} for the threshold block; the identity $c = 1 - \sum_k \partial_{\kappa_k} \psi$ holds to 10^{-15}. A Monte-Carlo check confirms `complete_surrogate()` samples the map it should ($|z| < 4$ at 4×10^5 draws), so the theory describes the implemented estimator and not merely the model.

The threshold-placement effect discussed in Section 13, at $e = 0$ under a logit:

κ	(0) binary	$(-0.5, 0.8)$	$(-1, 0.2, 1.3)$	(∓ 0.5)	(∓ 1.5)	(∓ 3)	(∓ 6)
$c(0)$	0.307	0.157	0.090	0.175	0.223	0.636	0.966

The last four entries are all $J = 3$: widening a three-category partition drives c from well below the binary value to nearly 1.

15.2 The variance decomposition (Theorems A1 and A2)

`R/theory_variance_terms.R` evaluates J_θ, σ_e^2, σ_u^2, G, Σ_β and C in closed form on the congenial testbed by product-grid quadrature, then compares Equation 12 against 4000 replications at $n = 20{,}000$ for $B \in \{1, 2, 5, 20\}$ and $\theta_0 \in \{1, 3\}$. All eight cells agree within 1.2 Monte-Carlo standard errors. Comparing across B is what identifies the individual terms; a single pooled variance ratio cannot, because its Monte-Carlo error is the same size as the 1 to 2 percent effect being measured, which is why the earlier ratio-based reading of these simulations was inconclusive rather than confirmatory.

The fixed-B spread of Equation 13 is confirmed directly: at $\theta_0 = 1$, $B = 20$, $n = 20{,}000$, the observed standard deviation of nT across replications is 2.13 against a mean of 9.73, while Equation 13 predicts $(1 + 1/B)V_{\mathrm{draw}}\sqrt{2/(B-1)} = 2.08$. The residual spread in the reported variance is therefore the between-draw sample variance, not estimation error, and it does not shrink with n.

Both closed forms behind these quantities rest on one identity, $p(1-p)(\mu_+ - \mu_-) = H(p)$ for the logistic law, which yields both $c(e) = 1 - H(F(e))$ and $C = \Sigma_\beta(A - G)$ with $A = E[zR]$. It is verified pointwise by quadrature to 8×10^{-16}.

The ordinal counterpart is checked by `R/theory_ordinal_variance_terms.R` on a correctly specified three-category cumulative-logit design with continuous treatment and exactly linear partialling nuisances. The calculation includes both threshold derivatives in G and matches the fixed-B sampling variance and expected Rubin variance within 2.3

Monte-Carlo standard errors in every cell at $n = 8{,}000$, 500 replications, $B \in \{1, 2, 5, 20\}$, and $\theta_0 \in \{1, 2.5\}$. At $B = 20$, the predicted T/V is 1.028 and 0.980, respectively. The ordinal threshold machinery therefore changes the same small Rubin discrepancy identified in the binary testbed but does not explain the much larger stage-5 gap.

`R/theory_ordinal_nuisance_ladder.R` then changes one partialling component at a time on the exact stage-5 DGP, using common samples, folds, imputation fits, parameter draws, and surrogates across arms. At $n = 2{,}000$, $B = 10$, and 100 replications, the current fixed-$\widehat{\ell}$ arm has empirical variance 0.01380 and mean $T = 0.02077$, while refitting $\widehat{\ell}$ on every draw has empirical variance 0.01375 and mean $T = 0.01295$. Thus the point estimator is unchanged but T/V falls from 1.505 to 0.942. The difference is almost entirely in $\bar{W}$, which falls from 0.01666 to 0.00876; the between-draw component is unchanged. This isolates reuse of the plug-in outcome nuisance as the source of the stage-5 variance inflation. `R/stage5r_ordinal_refit.R` performs the full correction check at $n = 2{,}000$, $B = 25$, and 200 replications. Bias is 0.016 (Monte-Carlo SE 0.007), coverage is 0.960 (Monte-Carlo SE 0.014), and $T/V = 1.214$ (Monte-Carlo SE 0.122), satisfying the stage's bias, coverage, and variance-calibration criteria. The corrected per-draw refit is therefore the ordinal default.

16 References

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